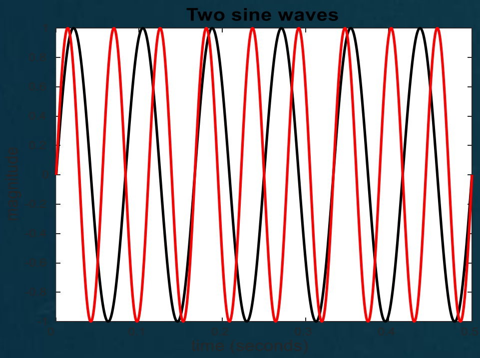
DATA QUALITY

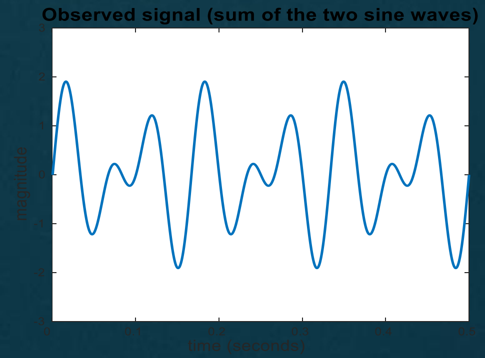
- It is unrealistic to expect that data will be perfect
- There may be problems due to human error, limitations of measuring devices, flows in the data collection process, application related issues

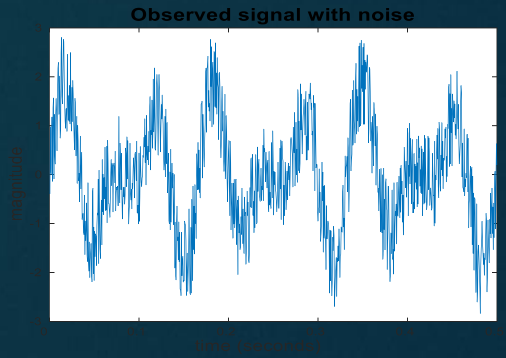
- Measurement and Data Collection Errors:
- Measurement Errors: Any problem resulting from the measurement process- value recorded differs from the true value to some extent
- Data Collection Errors: Omitting data objects/attribute values, inappropriately including a data object

Noise

- May involve the distortion of a value or the addition of spurious objects
- Often used with data that has spatial or temporal component
- For objects, noise is an extraneous object
- For attributes, noise refers to modification of original values
- Examples: distortion of a person's voice when talking on a phone
- The figures below show two sine waves of the same magnitude and different frequencies, the waves combined, and the two sine waves with random noise
- The magnitude and shape of the original signal is distorted

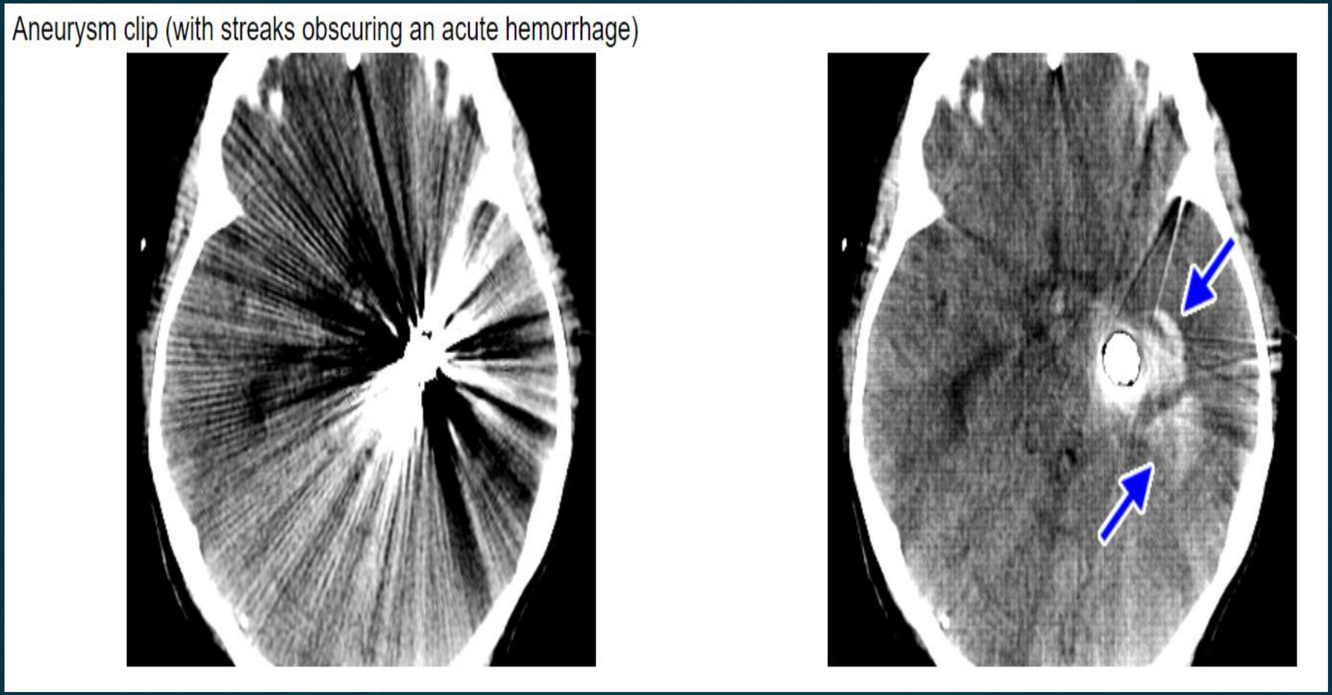






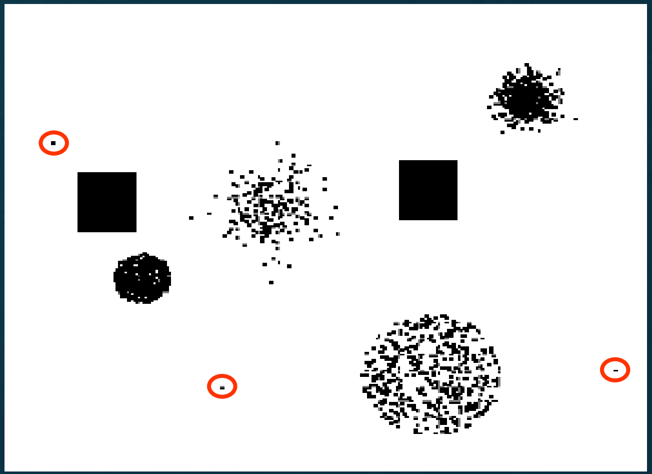
Artifact

- Deterministic distortions of the data
- Streak in the same place on a set of photographs, an image that becomes visibly distorted after compression



Outliers

- Outlier are either:
- i) Data objects with characteristics that are considerably different than most of the other data objects in the data set
- ii) Values of an attribute that are unusual with respect to the typical values for that attribute
- Outliers can be legitimate data objects
- Example: Fraud and network intrusion- Goal is to find unusual objects or events from among large number of normal ones



Missing Values

- Objects missing one or more attribute values
- Causes: Information not collected, attribute not applicable to data object
- There are strategies for dealing with missing data

i. Eliminate Data Objects or Attributes

- Simple and effective strategy is to eliminate objects with missing values
- Issues: If many objects have missing values, reliable analysis can be difficult if data objects are eliminated; also partially specified data will be lost
- If a dataset has only few objects, that have missing values, removing objects is not a good idea
- As an alternative, attributes that have missing values can be eliminated
- Should be done with caution, as attributes may be ones that are critical to the analysis

ii. Estimate Missing Values

- Sometimes, missing data can be reliably be estimated by using the remaining values
- If we consider a data set that ha many similar data points, attribute values of the points closest to the point with missing value are often used to estimate the missing value
- For continuous attribute, average attribute value of the nearest neighbor is used; for categorical attribute, most commonly occurring attribute value can be taken
- Example: for areas not containing a ground station, the precipitation can be estimated using values observed at nearby ground stations

iii. Ignore the Missing value during Analysis

- Many data mining approaches can be modified to ignore missing values
- Example: if objects are clustered, and the similarity between pairs of data objects to be calculated-
- If one or both objects of a pair have missing values, for some attributes, similarity can be calculated using only the attributes that do not have missing values

Inconsistent values

- Data can contain inconsistent values
- In address field, both zipcode and city are listed, but specified zipcode area is not contained in the city
- Some types of inconsistencies are easy to detect; in other cases, it may be necessary to consult an external source of information
- It is sometimes possible to correct the data; however, the correction of inconsistencies requires additional or redundant information

Duplicate Data

- A data set may include data objects that are duplicates, or almost duplicates of one another
- Two objects that actually represent a single object; data objects that are similar but not duplicates
- Deduplication: Process of dealing with duplication issues

Issues related to Applications

- Data quality issues can be considered from an application viewpoint
- Data is of high quality if it is suitable for its intended use

Timeliness:

- If data provides a snapshot of some ongoing phenomenon or process, it represents reality for only a limited time
- If data is out of date, then so are the methods and patterns that are based on it

Relevance:

- The available data must contain the information necessary for the application

Issues related to Applications

- Knowledge about the data:
- Ideally data sets are accompanied by documentation that describes different aspects of the data
- The quality of the documentation can either aid or hinder the subsequent analysis

Precision and Bias

- The quality of measurement process is often measured by precision and bias

Precision:

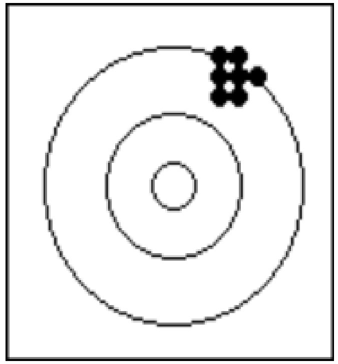
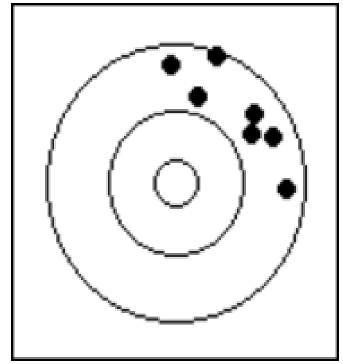
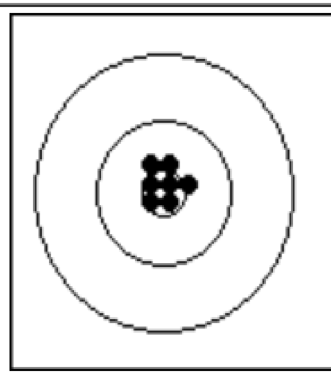
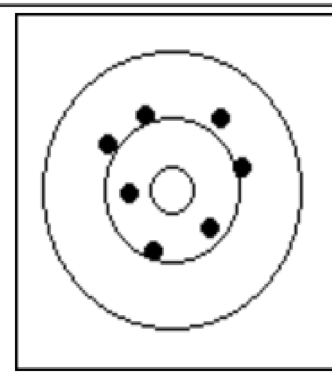
- The closeness of repeated measurements(of the same quantity) to one another
- Often measured by the standard deviation of a set of values

Bias:

- A systematic variation of measurements from the quantity being measured
- Measured by taking the difference between the mean of the set of values and the known value of the quantity being measured

Example:

- Consider a standard lab. weight with a mass of 1g
- Mass is measured 5 times and values be {1.015, 0.990, 1.013, 1.001,0.986}
- Mean=1.001
- Bias=0.001
- 0.013

	Precise	Imprecise
Biased		
Unbiased		

Accuracy

- The closeness of measurements to the true value of the quantity being measured